

# Abdelrahman Ahmed Al-Meshwady

Alexandria, Egypt | (+20)1116351047 | [abdelrahman.almeshwady@gmail.com](mailto:abdelrahman.almeshwady@gmail.com) |

[linkedin.com/in/abdelrahman-al-meshwady](https://www.linkedin.com/in/abdelrahman-al-meshwady) | [github.com/BodaTheDev](https://github.com/BodaTheDev) | Portfolio: [almeshwady.vercel.app](https://almeshwady.vercel.app)

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**Creative Software Engineer** who thrives at the intersection of robust backend architecture and exceptional product taste. Specializing in the modern TypeScript ecosystem (Node.js, React.js, Next.js), I engineer highly scalable, real-time infrastructure specifically to power fluid, low-latency, and meticulously crafted user experiences. Proven ability to own features end-to-end, from product ideation and architectural design to pixel-perfect UI implementation and deployment.

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## Work Experience

**Root Co. Software House | Alexandria, Egypt**

### Backend Developer & Cross-Functional Engineer | February 2025 – Present

- Serve as a core technical resource bridging the backend and frontend, accelerating feature delivery by implementing complex UI integrations in React/Next.js.
- Design and maintain scalable backend services using **Node.js**, leveraging **Express.js** for rapid prototyping and **NestJS** for maintainable enterprise applications.
- Engineer **RESTful APIs** with **TypeScript** to enforce strict type safety across the stack, reducing runtime errors and ensuring seamless data hydration for the frontend.
- Implemented authentication and secure integrations (JWT, RBAC), ensuring complex codebases remain secure and maintainable as the product scales.

### Front-End Developer (Part-Time) | October 2024 - September 2025

- Owned the **end-to-end** frontend development of web applications focused on logistics and e-commerce.
- Engineered intuitive dashboards and **real-time** interfaces, emphasizing state management, responsive layouts, and cross-browser **UI**.
- Integrated **high-performance** external services, including interactive Google Maps and Agora RTC, directly driving a 30% increase in user engagement through **lower latency** and better **UX**.

### Technical Project Manager | April 2022 - January 2025

- Acted as the technical product owner, making thoughtful product decisions to align stakeholder business goals with scalable engineering architecture.
  - Coordinated cross-functional teams using Agile methodologies, managing risk, scope, and technical delivery from initial idea to production.
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## Key Projects

### AI-Meshwady Portfolio Engine (Creative WebGL Development)

- Architected a high-performance 3D portfolio utilizing Three.js and custom GLSL shaders to deliver a cinematic, high-contrast "Villeneuve effect," merging deep technical execution with immersive visual storytelling.
- Engineered a reactive, physical lighting rig (Arancio Xanto) that tracks cursor movement using a precise 0.01 Lerp-based damping function, ensuring the UI delivers a heavy, tactile feeling of digital mass rather than a weightless web experience.
- Implemented complex kinetic typography and variable-font interactions (GSAP) while maintaining strict architectural performance, achieving perfect 100/100 Lighthouse desktop scores.

### WeCare App (IoT & Real-time Tracking)

- Designed a containerized microservices architecture (Docker, MQTT, Redis) to enable interactive, live trip monitoring on the frontend.
- Identified a critical architectural bottleneck threatening business runway; migrated the real-time tracking module from Python (Django) to Node.js, scaling WebSocket concurrency to 1,000+ users while radically dropping server costs.
- Implemented secure, self-hosted live video streaming using WebRTC and HLS for ultra-low latency playback.

### Relive App (Telehealth & RTC)

- Built a modular, production-ready Node.js backend for secure multi-role authentication and API delivery.
  - Engineered the video consultation feature end-to-end, integrating the Agora Web SDK on the React frontend for seamless, low-latency real-time communication.
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## Technical Skills

- **Creative Development & UI:** React.js, Next.js, Three.js, GSAP, WebGL, SCSS, Styled-components, Material-UI, Bootstrap, Framer Motion, Accessibility (a11y).
  - **Design & Prototyping:** Figma, Adobe Photoshop, Illustrator, Adobe XD, Responsive Interface Design.
  - **Backend & Core Architecture:** TypeScript, JavaScript (Node.js), NestJS, Express.js, Angular, Python, Django, RESTful API Design, JWT Authentication, WebSockets, MQTT.
  - **Databases & ORMs:** PostgreSQL, MongoDB, Redis, TypeORM, Prisma, SQL.
  - **DevOps & Infrastructure:** Docker, Nginx, PM2, Ubuntu Server, AWS, Vercel, SSL/DNS Management, CI/CD Pipelines.
  - **Tools & Methodologies:** Git, GitHub, Agile (Scrum, Kanban), Jira, ClickUp, Trello, Agora Web SDK.
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## Education

- Bachelor of Commerce, Accounting (English Dept.) | Alexandria University, Graduated 2022